

DSA0602-Data Handling and Visualization

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EXPERIMENT 1: Monthly Sales Analysis

AIM

To visualize monthly sales data using different charts in R Programming.

PROCEDURE

Open RStudio.

Load the required packages. Create the monthly sales dataset.

Generate line chart, bar chart, scatter plot and interactive dashboard. Execute the program and observe the output.

PROGRAM

Step 1: Load Package library(ggplot2) library(plotly)

Step 2: Create Dataset

```
sales <- data.frame(  
  Month = c("Jan", "Feb", "Mar", "Apr", "May"), Sales = c(1200, 1500, 1800, 1600, 2000)  
)
```

```
print(sales)
```

Step 3: Line Chart

```
ggplot(sales, aes(x = Month, y = Sales, group = 1)) + geom_line(color = "blue", linewidth =  
  1.2) + geom_point(color = "red", size = 3) +  
  labs(  
    title = "Monthly Sales Trend", x = "Month",  
    y = "Sales"  
  ) +  
  theme_minimal()
```

Step 4: Bar Chart

```
ggplot(sales, aes(x = Month, y = Sales, fill = Month)) + geom_bar(stat = "identity") +
```

```
labs(
  title = "Monthly Sales Comparison", x = "Month",
  y = "Sales"
) +
theme_minimal()
```

Step 5: Scatter Plot

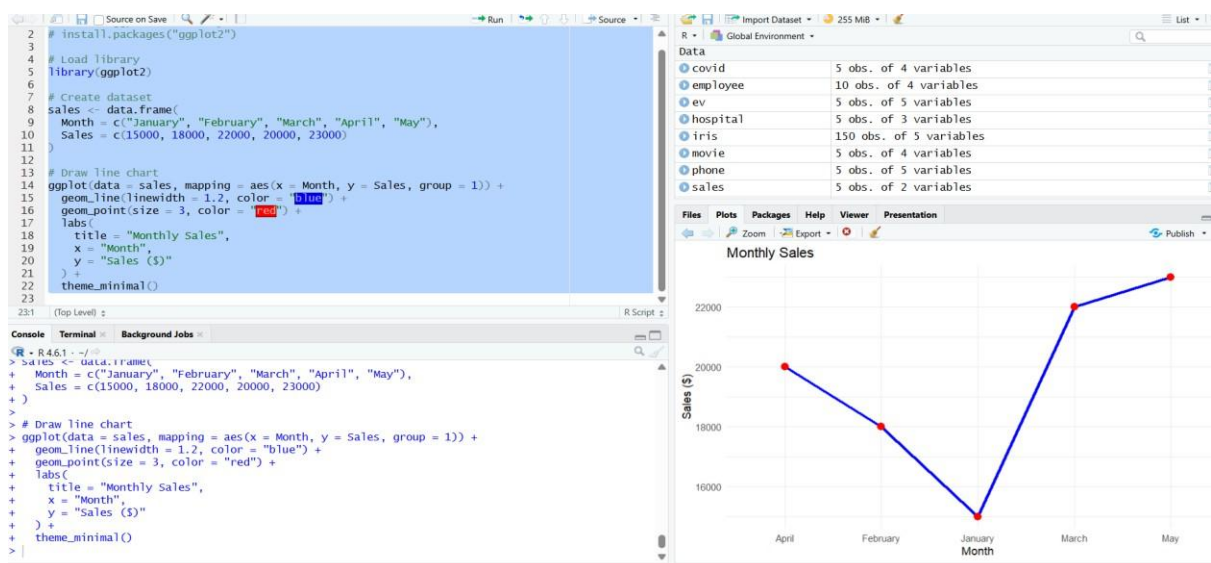
```
ggplot(sales, aes(x = Month, y = Sales)) + geom_point(size = 4, color = "blue") +
  geom_smooth(aes(group = 1), method = "lm", se = FALSE, color = "red") + labs(
    title = "Monthly Sales Scatter Plot", x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```

Step 6: Interactive Dashboard

```
p1 <- plot_ly(sales,
  x = ~Month, y = ~Sales,
  type = "scatter",
  mode = "lines+markers")

p2 <- plot_ly(sales,
  x = ~Month, y = ~Sales, type = "bar")

subplot(p1, p2, nrows = 2)
```



RESULT

The monthly sales data was successfully visualized using line chart, bar chart, scatter plot and an interactive dashboard.

EXPERIMENT 2: Customer Feedback Analysis

AIM

To visualize customer feedback data using different charts in R Programming.

PROCEDURE

Open RStudio.

Load the required packages. Create the customer dataset.

Generate histogram, pie chart, stacked bar chart and word cloud. Execute the program and observe the output.

PROGRAM

Step 1: Load Packages library(ggplot2) library(wordcloud)

library(RColorBrewer) **Step 2: Create Dataset** customer <- data.frame(

Age = c(20,25,30,35,40),

Satisfaction = c("High","Medium","High","Low","Medium")

)

print(customer)

Step 3: Histogram ggplot(customer, aes(x = Age)) + geom_histogram(binwidth = 5,
fill = "skyblue", color = "black") +

labs(

title = "Customer Age Distribution", x = "Age",

y = "Frequency"

) +

theme_minimal() **Step 4: Pie Chart** pie(

table(customer\$Satisfaction), col = rainbow(3),

main = "Customer Satisfaction Level"

)

Step 5: Stacked Bar Chart

customer\$Age_Group <- c("20-25","20-25","30-35","30-35","40+")

ggplot(customer,

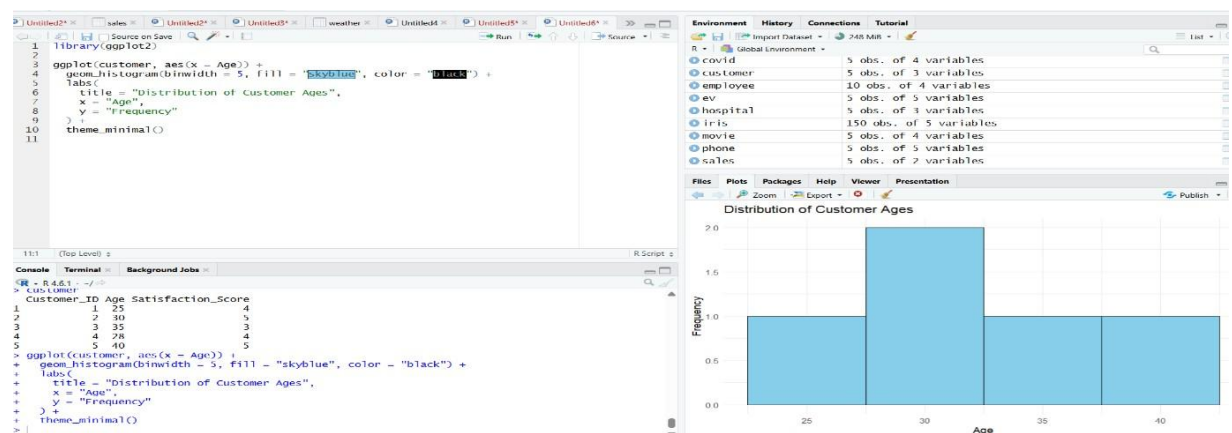
aes(x = Age_Group, fill = Satisfaction)) +

geom_bar() +

```

labs(
  title = "Satisfaction by Age Group", x = "Age Group",
  y = "Count"
) +
theme_minimal() Step 6: Word Cloud wordcloud(
words = customer$Satisfaction, freq = c(2,2,1,1,2),
colors = brewer.pal(8,"Dark2")
)

```



RESULT

The customer feedback data was successfully visualized using histogram, pie chart, stacked bar chart and word cloud.

These versions are **medium-sized**, clean, and suitable for submission in a lab record.

EXPERIMENT 3: Employee Performance Analysis

AIM

To analyze and visualize employee performance data using R Programming.

PROCEDURE

1. Open RStudio.
2. Load the required packages.
3. Create the employee dataset.
4. Generate line chart, bar chart, scatter plot and interactive dashboard.
5. Execute the program and observe the output.

PROGRAM

Step 1: Load Packages library(ggplot2) library(plotly)

Step 2: Create Dataset employee <- data.frame(Employee_ID = c(1,2,3,4,5),
Department = c("Sales","HR","Marketing","Sales","HR"), Years_of_Service = c(5,3,7,4,2),
Performance_Score = c(85,92,78,90,76)
)

print(employee)

Step 3: Line Chart

ggplot(employee,
aes(x = Years_of_Service, y = Performance_Score, group = 1)) +
geom_line(color = "blue", linewidth = 1.2) + geom_point(color = "red", size = 3) +
labs(
title = "Employee Performance Trend", x = "Years of Service",
y = "Performance Score"
) +

theme_minimal() **Step 4: Bar Chart** ggplot(employee,

aes(x = Department, fill = Department)) +
geom_bar() + labs(
title = "Employees by Department", x = "Department",
y = "Number of Employees"
) +

theme_minimal() **Step 5: Scatter Plot** ggplot(employee,

aes(x = Years_of_Service,

```

    y = Performance_Score)) + geom_point(size = 4, color = "blue") + geom_smooth(method
= "lm",
    se = FALSE,
    color = "red") +
labs(
  title = "Years of Service vs Performance", x = "Years of Service",
  y = "Performance Score"
) +
theme_minimal()

```

Step 6: Interactive Dashboard

```

p1 <- plot_ly(employee,
  x = ~Years_of_Service,
  y = ~Performance_Score, type = "scatter",
  mode = "lines+markers")

```

```

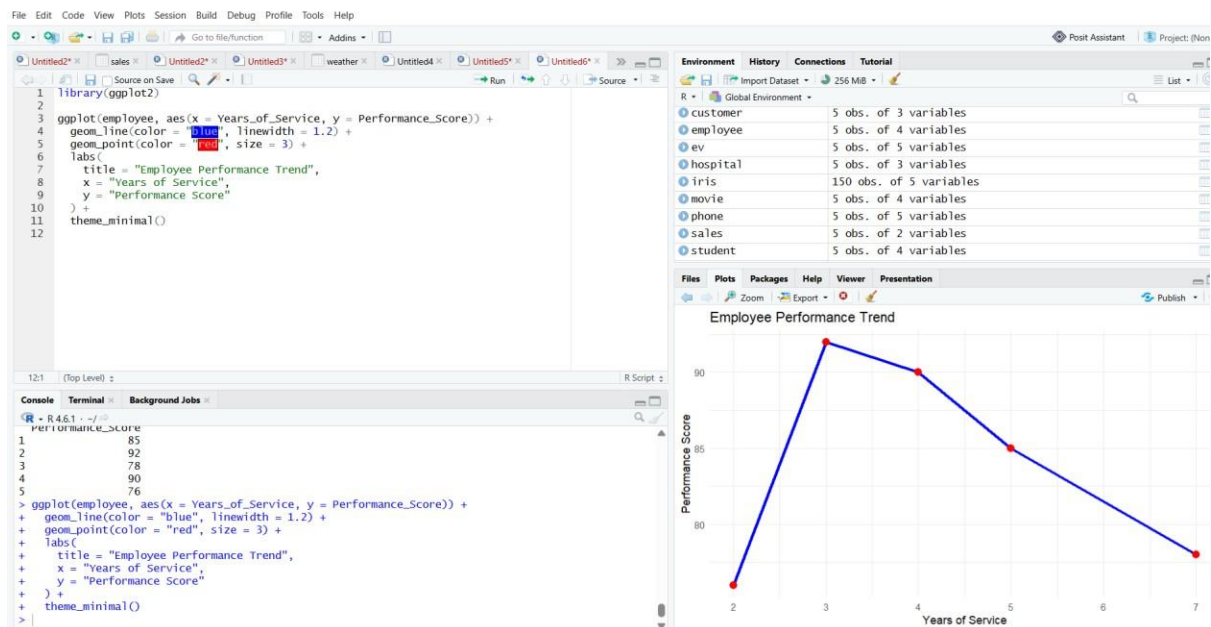
p2 <- plot_ly(employee,
  x = ~Department, type = "bar")

```

```

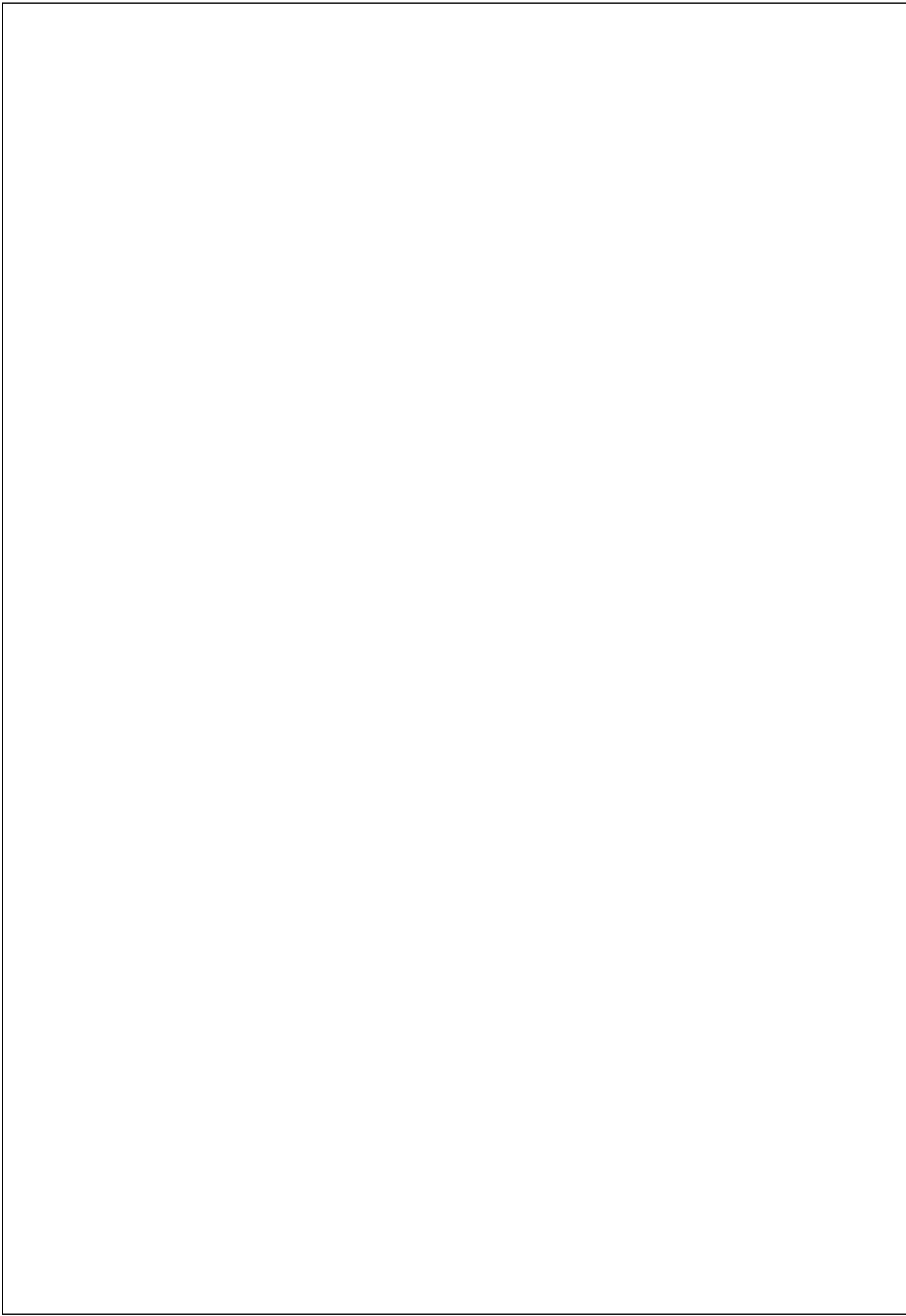
subplot(p1, p2, nrow = 2)

```



RESULT

Employee performance data was successfully analyzed and visualized using line chart, bar chart, scatter plot and interactive dashboard.



EXPERIMENT 4: Product Inventory Analysis

AIM

To visualize product inventory data using different charts in R Programming.

PROCEDURE

1. Open RStudio.
2. Load the required packages.
3. Create the inventory dataset.
4. Generate bar chart, stacked bar chart, scatter plot and interactive dashboard.
5. Execute the program and observe the output.

PROGRAM

Step 1: Load Packages library(ggplot2) library(plotly)

Step 2: Create Dataset

```
inventory <- data.frame(  
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"), Category =  
  c("Electronics", "Electronics", "Furniture", "Furniture", "Furniture"),  
  Price = c(500, 650, 450, 700, 550), Quantity = c(250, 175, 300, 200, 220)  
)
```

```
print(inventory)
```

Step 3: Bar Chart

```
ggplot(inventory,  
  aes(x = Product_Name, y = Quantity,  
    fill = Product_Name)) + geom_bar(stat = "identity") + labs(  
  title = "Product Inventory", x = "Product Name",  
  y = "Quantity"  
) +  
theme_minimal()
```

Step 4: Stacked Bar Chart

```
ggplot(inventory, aes(x = Category,  
  y = Quantity,  
  fill = Product_Name)) + geom_bar(stat = "identity") + labs(  
  title = "Inventory by Category", x = "Category",  
  y = "Quantity"
```

```
) +
```

```
theme_minimal()
```

Step 5: Scatter Plot

```
ggplot(inventory, aes(x = Price,
```

```
  y = Quantity)) + geom_point(size = 4, color = "blue") + geom_smooth(method = "lm",
```

```
  se = FALSE,
```

```
  color = "red") +
```

```
labs(
```

```
  title = "Price vs Quantity", x = "Price",
```

```
  y = "Quantity"
```

```
) +
```

```
theme_minimal()
```

Step 6: Interactive Dashboard

```
p1 <- plot_ly(inventory,
```

```
  x = ~Product_Name, y = ~Quantity,
```

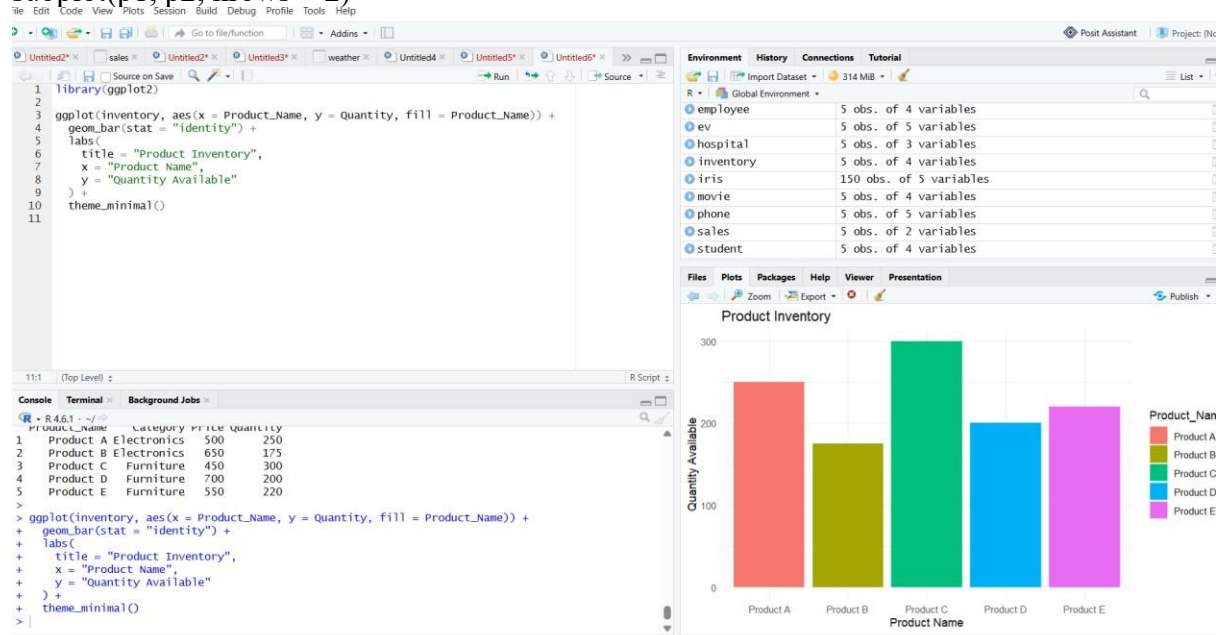
```
  type = "bar")
```

```
p2 <- plot_ly(inventory,
```

```
  x = ~Price,
```

```
  y = ~Quantity, type = "scatter", mode = "markers")
```

```
subplot(p1, p2, nrow = 2)
```



RESULT

Product inventory data was successfully visualized using bar chart, stacked bar chart, scatter plot and interactive dashboard.

These are in the same format and level of detail as Experiments 1 and 2, making all four experiments consistent for your Word lab record.

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QUESTION 5: Website Analytics

AIM

To analyze website traffic data by creating different visualizations using R Programming.

PROCEDURE

1. Open RStudio.
2. Load the required libraries.
3. Create the website traffic dataset.
4. Generate a line chart.
5. Generate a bar chart.
6. Create a stacked area chart.
7. Build an interactive dashboard.
8. Execute the program and observe the output.

PROGRAM

Step 1: Load Packages `library(ggplot2) library(plotly) library(tidyr)`

Step 2: Create Dataset

```
traffic <- data.frame(
```

```
  Date = as.Date(c( "2023-01-01",
```

```
    "2023-01-02",
```

```
    "2023-01-03",
```

```
    "2023-01-04",
```

```
    "2023-01-05"
```

```
)),
```

```
  Page_Views = c( 1500,
```

```
    1600,
```

```
    1400,
```

```
1650,  
1800  
,  
  
CTR = c( 2.3,  
2.7,  
2.0,  
2.4,  
2.6  
)  
)
```

```
print(traffic)
```

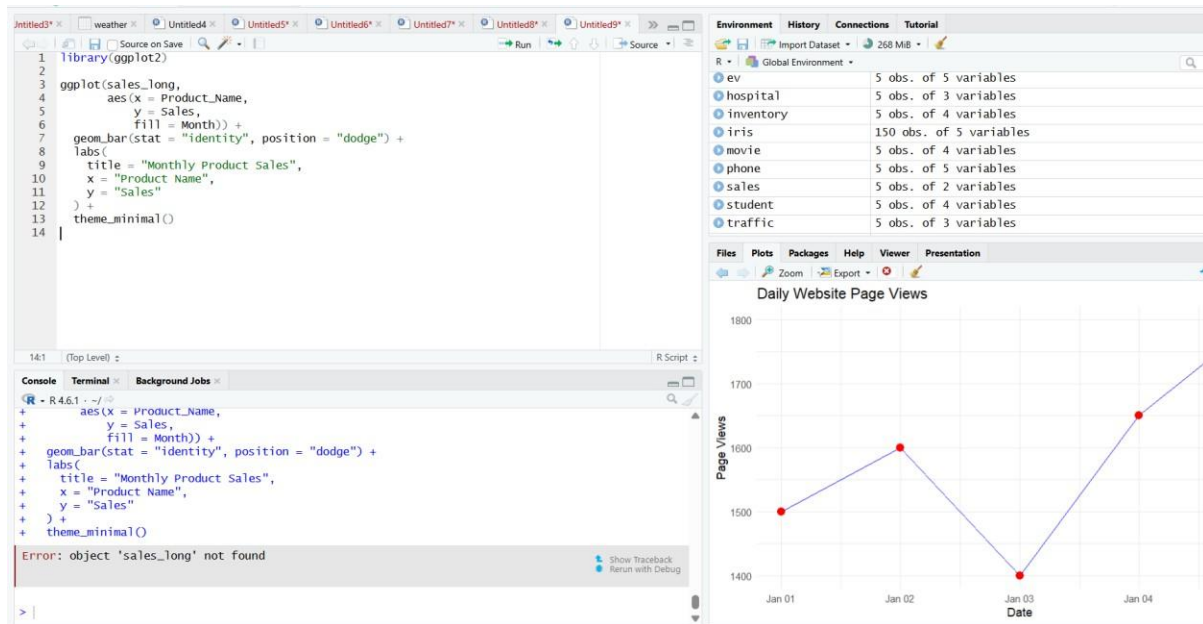
Question 1 Line Chart `ggplot(traffic, aes(x = Date,
y = Page_Views, group = 1)) +`

```
geom_line(color = "blue", linewidth = 1.2) +
```

```
geom_point(color = "red", size = 3) +
```

```
labs(  
title = "Daily Website Page Views", x = "Date",  
y = "Page Views"  
) +
```

```
theme_minimal()
```



RESULT

The website traffic data was successfully visualized using a line chart, bar chart, stacked area chart, and an interactive Plotly dashboard in RStudio.